

LARGE INTESTINE

Regular Exam Notes - Caecum, Appendix and Colon

At a Glance

- Extent: from ileocaecal junction to anus.
- Length: about 1.5 m in the adult.
- Parts: caecum with appendix, colon, rectum and anal canal.
- Main role: absorption of water/salts and formation, storage and expulsion of feces.

Exam Focus

- Cardinal features: taeniae coli, appendices epiploicae and haustra.
- Caecum and appendix: relations, surface marking, positions and blood supply.
- Colon: subdivisions, flexures, arterial supply and lymphatic drainage.
- Applied: appendicitis, carcinoma, diverticulosis, volvulus and Hirschsprung disease.

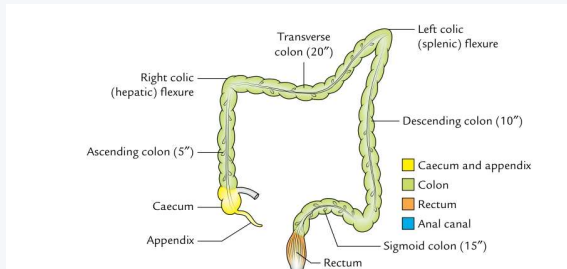


Fig. 10.7 - Parts of the large intestine.

How to read this diagram

The large intestine begins at the caecum, receives the ileum at the ileocaecal junction, and ends at the anal canal. It frames the coils of small intestine like an arch.

The colon is divided into ascending, transverse, descending and sigmoid parts. The right colic flexure lies under the liver, while the left colic flexure lies near the spleen.

Remember: caecum and colon show haustrations and taeniae coli, but rectum and anal canal do not show appendices epiploicae.

DEFINITION, PARTS AND FUNCTIONS

Definition

The large intestine is the terminal part of the digestive tube. It begins at the ileocaecal junction as the caecum and terminates at the anus. It is shorter than the small intestine but wider in diameter and has characteristic surface features.

Parts

- Caecum and vermiform appendix.
- Colon: ascending, transverse, descending and sigmoid colon.
- Rectum.
- Anal canal.

Functions

- Absorption of water and electrolytes from intestinal contents.
- Formation, lubrication and expulsion of feces.
- Storage of fecal matter before defecation.
- Bacterial flora synthesize vitamin B complex and support gut immunity.

STRUCTURE OF THE WALL

Layers of the colon

- Mucosa: no villi and no permanent plicae circulares; epithelium is absorptive with many goblet cells in long crypts of Lieberkuhn.
- Submucosa: loose areolar tissue containing blood vessels, lymph vessels and Meissner plexus.
- Muscle layer: inner circular muscle and outer longitudinal muscle arranged into three bands called taeniae coli.
- Serosa/adventitia: transverse and sigmoid colon have serosa; ascending and descending colon are partly covered by adventitia posteriorly.

Muscle Layer

It consists of outer longitudinal and inner circular layers of smooth muscle. The outer longitudinal layer is not

Teniae Coli

These are three ribbon-like bands of the longitudinal muscle coat. These bands converge proximally at the base of

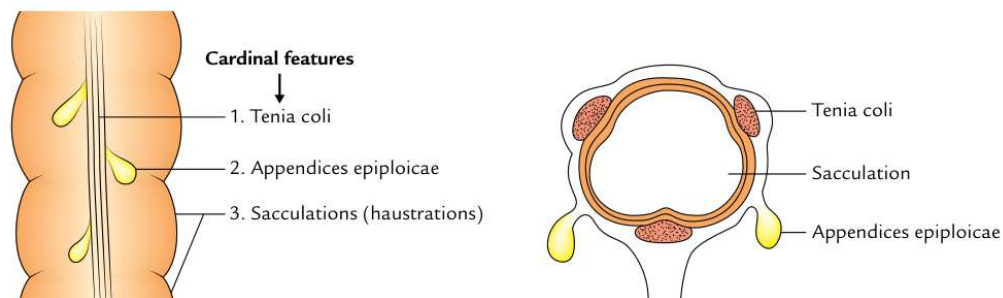


Fig. 10.8 - Cardinal features of the large intestine.

The three identifying features are taeniae coli, appendices epiploicae and sacculations or haustra. Taeniae coli are shorter than the gut wall, so the wall becomes puckered into haustra. Appendices epiploicae are peritoneal fat tags attached to the colon.

Table: Small intestine vs large intestine

Feature	Small intestine	Large intestine
Length	About 6 m	About 1.5 m
Lumen	Narrower	Wider
Mobility	More mobile	Less mobile overall
Villi	Present	Absent
Peyer patches	Present, especially ileum	Absent
Appendices epiploicae	Absent	Present except appendix, rectum and anal canal
Taeniae coli	Absent	Present in caecum and colon
Sacculations/haustra	Absent	Present

CAECUM

Definition and position

The caecum is the large dilated blind sac at the commencement of the large intestine. It is situated in the right iliac fossa above the lateral half of the inguinal ligament. It communicates superiorly with the ascending colon, medially with the ileum and posteromedially with the appendix.

Dimensions and shape

- Length: about 6 cm.
- Width: about 7.5 cm.
- In adults it usually appears as a dilated pendulous sac below the ileocaecal junction.
- The base of appendix is attached to the posteromedial wall of the caecum.

Clinical point

- The caecum acts as a useful guide during surgery for intestinal obstruction.
- If the caecum is distended, obstruction is likely in the large intestine.
- If the caecum is empty, obstruction is more likely in the small intestine.

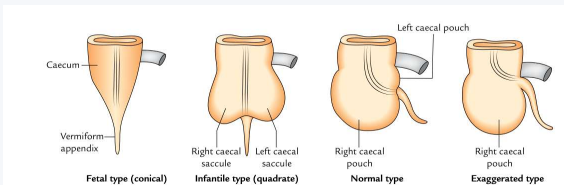


Fig. 10.9 - Types of caecum.

Types of caecum

Four types are described according to adult caecal development: fetal/conical, infantile/quadrate, normal and exaggerated types.

In the normal type, the right caecal pouch is larger and the appendix is attached on the posteromedial aspect near the ileocaecal junction.

This explains why the base of the appendix is found by following the three taeniae coli to their convergence.

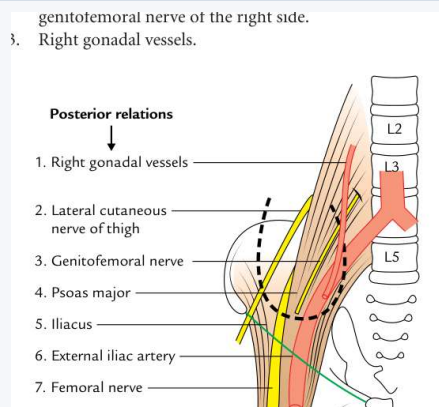


Fig. 10.10 - Posterior relations of caecum.

Relations of caecum

Anteriorly, the caecum is related to coils of small intestine, greater omentum and anterior abdominal wall in the right iliac region.

Posteriorly, it is related to right psoas major, iliacus, femoral nerve, lateral cutaneous nerve of thigh, genitofemoral nerve, right gonadal vessels and sometimes the right external iliac artery.

The retrocaecal recess often contains the vermiform appendix, making retrocaecal appendicitis clinically important.

PERITONEAL RELATIONS AND INTERIOR OF CAECUM

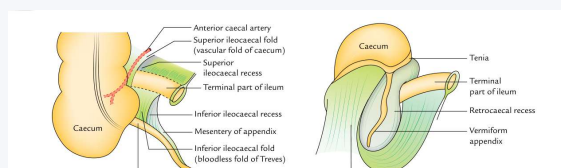


Fig. 10.11 - Peritoneal relations of caecum.

Peritoneal folds and recesses

In most individuals the caecum is almost completely covered by peritoneum and possesses a retrocaecal recess.

Important folds are superior ileocaecal fold, inferior ileocaecal fold and caecal fold. The inferior ileocaecal fold is also called the bloodless fold of Treves.

These folds form ileocaecal recesses. A recess may

lodge the appendix or become a site where internal herniation is possible.

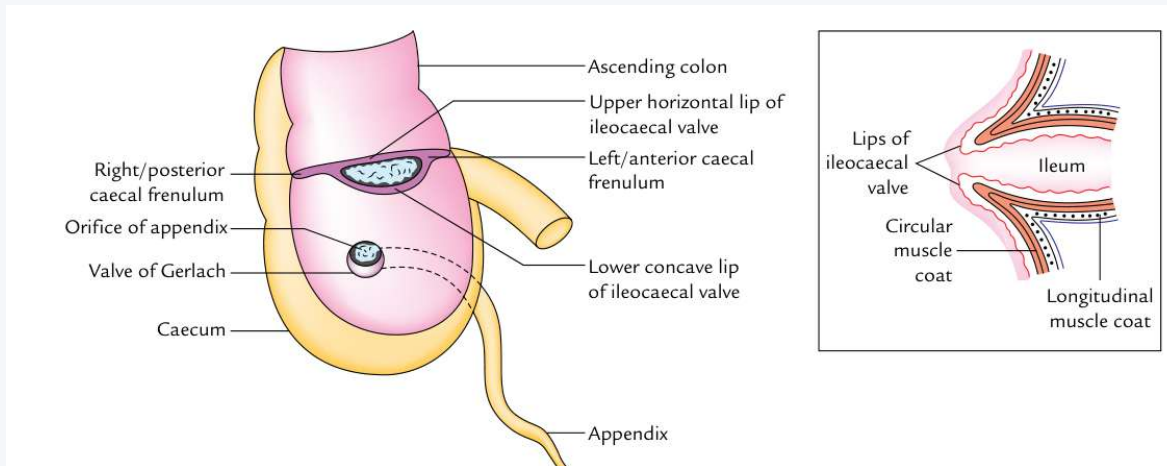


Fig. 10.12 - Interior of caecum with ileocaecal and appendicular orifices.

The ileocaecal orifice lies on the posteromedial aspect at the caecocolic junction and is guarded by the ileocaecal valve. The appendicular orifice lies about 2 cm below and slightly behind the ileocaecal orifice and is guarded by the valve of Gerlach.

Ileocaecal valve

- Has upper and lower lips formed by circular muscle thickening with mucosa and submucosa.
- Regulates passage from ileum into caecum.
- Prevents regurgitation from caecum back into ileum.

Appendicular orifice

- Small circular opening, about 2 cm below and slightly behind the ileocaecal orifice.
- Guarded by an insignificant mucosal fold called valve of Gerlach.
- Explains why obstruction of the appendicular lumen can produce appendicitis.

Caecum - quick exam tip

In an exam answer, write the caecum as a blind sac in the right iliac fossa receiving the ileum and giving origin to the appendix. Add the ileocaecal valve, appendicular orifice and peritoneal recesses as high-yield points. Clinically, the caecum is important in appendicitis, intestinal obstruction and surgical identification of the ileocaecal region.

VERMIFORM APPENDIX

Definition and parts

The vermiform appendix is a narrow worm-like diverticulum arising from the posteromedial wall of the caecum about 2 cm below the ileocaecal junction. It has three parts: base, body and tip. The base is fixed; the body and tip may occupy different positions.

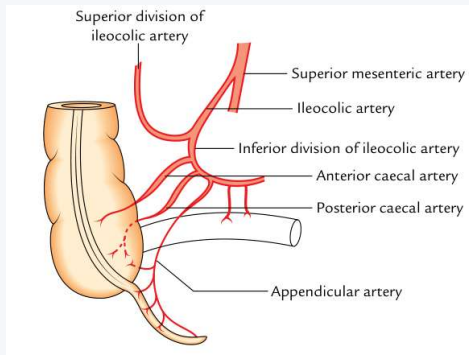


Fig. 10.13 - Arterial supply of caecum and appendix.

Blood and lymph supply

The caecum is supplied by anterior and posterior caecal branches of the ileocolic artery, a branch of the superior mesenteric artery.

The appendix is supplied by a single appendicular artery, usually a branch of the inferior division of the ileocolic artery. This artery runs in the free margin of mesoappendix.

Venous drainage follows the arteries to the superior mesenteric vein. Lymph drains to ileocolic nodes and then to superior mesenteric nodes.

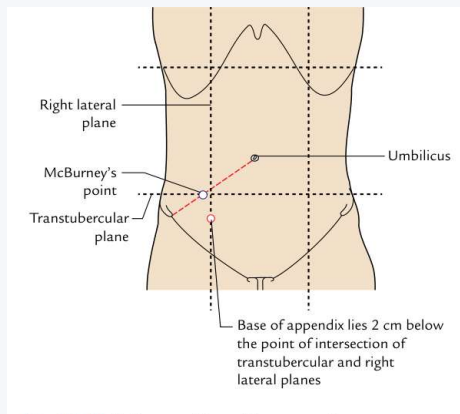


Fig. 10.14 - Surface marking of appendix.

McBurney point

The base of appendix is marked about 2 cm below the intersection of the transtubercular plane and right lateral plane.

Clinically, this point closely corresponds to McBurney point, commonly used for locating maximum tenderness in acute appendicitis.

McBurney point is classically at the junction of lateral one-third and medial two-third of a line from the right anterior superior iliac spine to the umbilicus.

Position quick list

- Retrocaecal/retrocolic: commonest position.
- Pelvic: second common position and may irritate pelvic organs.
- Pre-ileal and post-ileal: related to terminal ileum.
- Subcaecal or midinguinal: may point towards inguinal ligament.

Why position matters

- Different positions alter the site of tenderness and symptoms.
- Retrocaecal appendix may not produce marked guarding early.
- Pelvic appendix may cause urinary frequency or rectal irritation.
- Pre-ileal inflammation may spread easily in peritoneal cavity.

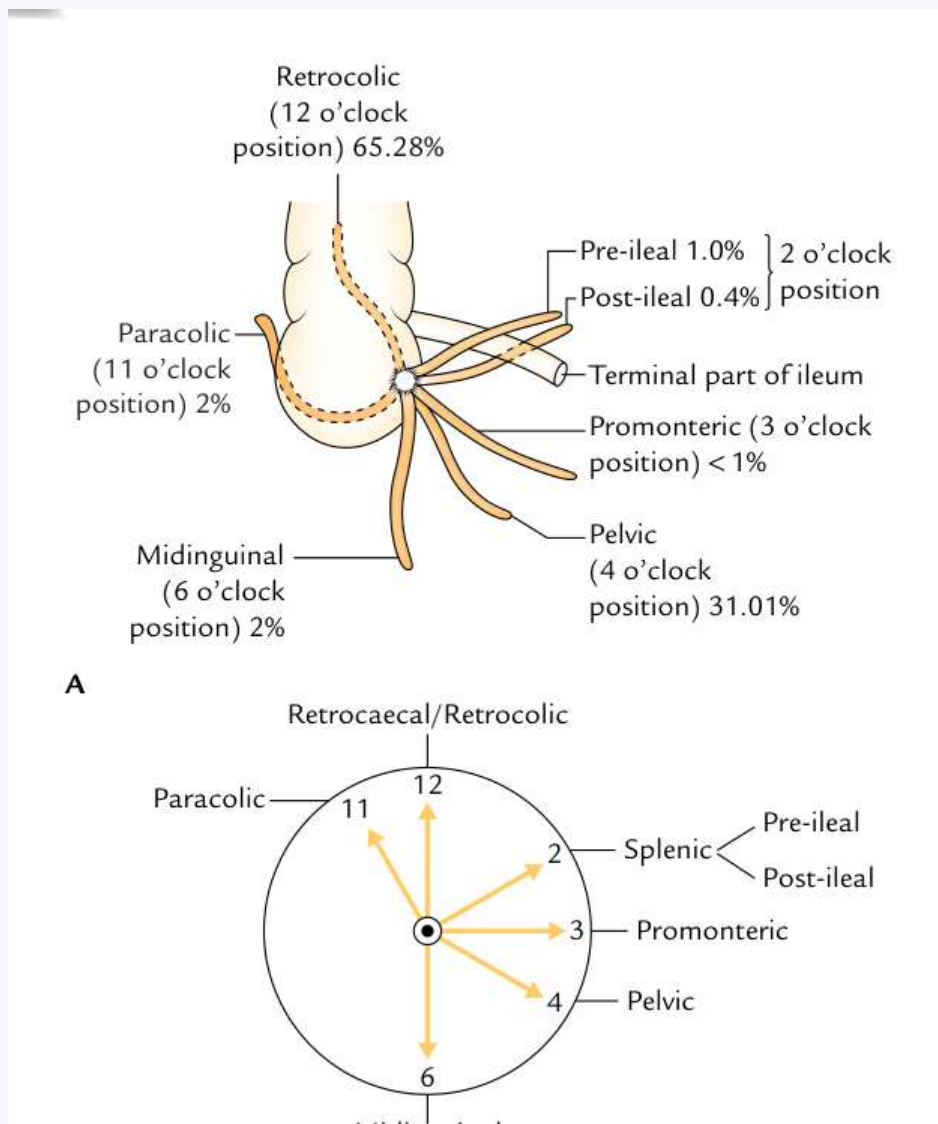


Fig. 10.15 - Positions of vermiform appendix.

The base is fixed but the rest of appendix may lie in several directions. Retrocaecal/retrocolic position is the commonest. Pelvic position is the second common position. Pre-ileal position is clinically dangerous because inflammation can spread easily into the general peritoneal cavity.

Nerve supply of appendix

- Sympathetic pain fibres arise from T10 segment through superior mesenteric plexus.
- Early appendicular pain is referred to the umbilical region because appendix is a midgut derivative.
- Later pain localizes to right iliac fossa when parietal peritoneum becomes inflamed.

Applied anatomy

- Appendicular artery is an end artery; thrombosis may lead to gangrene and perforation.
- Retrocaecal appendicitis may produce psoas irritation and a positive psoas test.
- Pelvic appendix may irritate bladder or rectum causing urinary or rectal symptoms.

COLON AND COLIC FLEXURES

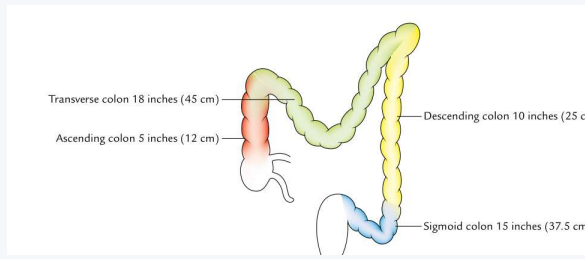


Fig. 10.18 - Subdivisions and length of colon.

Colon overview

Ascending colon is about 12.5 cm and runs from caecum to the right colic flexure under the liver.

Transverse colon is the longest and most mobile part of the colon. It forms a U-shaped dependent loop between the two colic flexures.

Descending colon runs down the left side to the left iliac fossa. Sigmoid colon is an S-shaped mobile loop with sigmoid mesocolon.

Table: Right and left colic flexures

Feature	Right colic flexure	Left colic flexure
Region	Right lumbar region, related to right lobe of liver	Left hypochondrium, below the spleen
Level	About L2; below transpyloric plane	About T12; above transpyloric plane
Angulation	Wider angle	More acute angle
Attachment	No attachment to diaphragm	Attached by phrenicocolic ligament
Mobility	Relatively more mobile	Relatively fixed and higher

Ascending colon

- Continuation of caecum to inferior surface of liver.
- Mostly retroperitoneal; covered by peritoneum anteriorly and laterally.
- Posteriorly related to iliacus, quadratus lumborum and transversus abdominis.
- Crosses lateral cutaneous nerve of thigh, ilioinguinal and iliohypogastric nerves.

Transverse colon

- Longest and most mobile part of the large intestine.
- Suspended by transverse mesocolon.
- Anterior relation: greater omentum and anterior abdominal wall.
- Posterior relation: second part of duodenum, head of pancreas and small intestinal coils.

Descending colon

- Extends from left colic flexure to pelvic brim.
- Mostly retroperitoneal and lies in the left paracolic gutter.
- Posteriorly related to left kidney region, iliacus, quadratus lumborum and abdominal wall muscles.
- Continues as sigmoid colon in the left iliac fossa.

Sigmoid colon

- Mobile S-shaped pelvic part of colon.
- Suspended by sigmoid mesocolon with inverted V-shaped root.
- Common site for diverticulosis and volvulus due to mobility.
- Continues as rectum in front of S3 vertebra.

BLOOD SUPPLY, VENOUS AND LYMPHATIC DRAINAGE

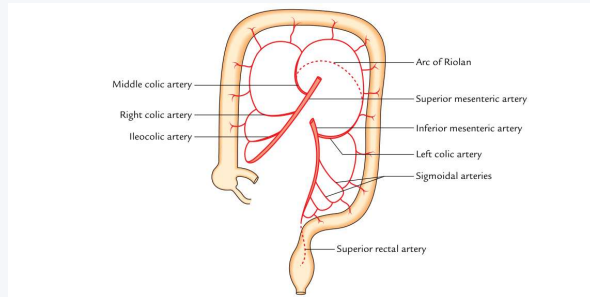


Fig. 10.19 - Arterial supply of colon and marginal artery.

Arterial pattern

The colon receives branches from both superior and inferior mesenteric arteries.

Ileocolic, right colic and middle colic arteries supply the caecum, ascending colon and proximal two-third of transverse colon.

Left colic, sigmoid arteries and superior rectal artery supply distal one-third of transverse colon, descending colon, sigmoid colon and upper rectum.

The marginal artery of Drummond forms a continuous anastomotic channel along the colon.

Critical point of colon

- Located at the splenic flexure.
- Here the ascending branch of left colic may fail to anastomose adequately with the left branch of middle colic.
- Poor anastomosis can make the splenic flexure vulnerable to ischemia.

Venous drainage

- Veins accompany the arteries.
- Ileocolic, right colic and middle colic veins drain into superior mesenteric vein.
- Left colic and sigmoid veins drain into inferior mesenteric vein.
- Both systems ultimately drain into portal circulation.

Arterial territories summary

- SMA territory: caecum, appendix, ascending colon and proximal two-third of transverse colon.
- IMA territory: distal one-third of transverse colon, descending colon, sigmoid colon and upper rectum.
- The marginal artery runs close to the inner margin of colon and provides collateral circulation between colic branches.
- Watershed areas are vulnerable when collateral anastomoses are poor.

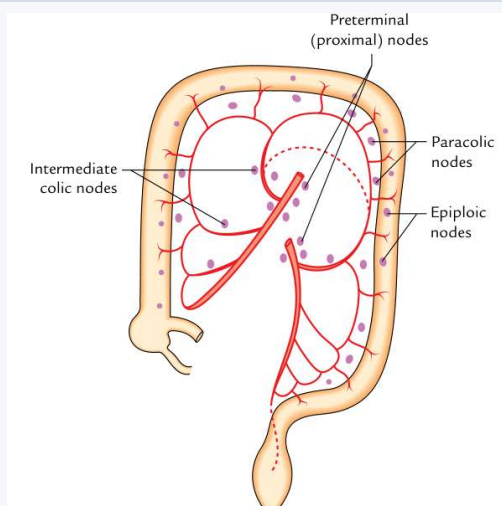


Fig. 10.20 - Lymphatic drainage of colon.

Lymphatic pattern

Lymphatic drainage is clinically important because carcinoma of colon spreads through lymphatics.

Epiploic nodes lie on the wall of colon; paracolic nodes lie along the marginal artery; intermediate colic nodes lie along colic arteries; preterminal nodes lie along the main mesenteric vessels.

During cancer surgery, lymph nodes and the arterial territory of the affected segment are removed together.

Nerve supply of colon

The proximal colon up to the proximal two-third of transverse colon is derived from midgut and supplied mainly via superior mesenteric plexus with vagal parasympathetic fibres. The distal one-third of transverse colon, descending colon and sigmoid colon are hindgut derivatives; their parasympathetic supply comes from pelvic splanchnic nerves (S2-S4).

- Sympathetic fibres reduce motility and carry visceral pain.
- Parasympathetic fibres increase peristalsis and secretion.
- Pain from midgut is referred around the umbilicus; pain from hindgut tends to be lower abdominal/suprapubic.

APPLIED ANATOMY

Appendicitis

- Initial pain is around umbilicus due to visceral afferents from T10 segment.
- Pain later localizes to right iliac fossa when parietal peritoneum is involved.
- Maximum tenderness commonly occurs near McBurney point.
- Retrocaecal appendix may give positive psoas test.

Carcinoma of colon

- Common in older age groups and spreads mainly through lymphatics and portal venous route.
- May cause obstruction, bleeding, altered bowel habit or weight loss.
- Right-sided lesions may present with anemia; left-sided lesions more often cause obstruction.
- Surgical removal follows arterial and lymphatic territories.

Diverticulosis and diverticulitis

- Diverticula are mucosal herniations through weak areas of muscle wall, commonly in sigmoid colon.
- They occur where vessels pierce the circular muscle layer.
- Inflammation of diverticula is called diverticulitis and may cause pain, fever or perforation.

Volvulus and intussusception

- Volvulus is twisting of gut around the axis of mesentery; sigmoid colon is commonly involved.
- It may obstruct lumen and compromise blood supply.
- Intussusception is invagination of proximal bowel into distal bowel and may cut off vascular supply.

Hirschsprung disease

- Congenital absence of myenteric plexus in distal colon/rectum.
- Affected segment remains contracted and proximal colon becomes dilated.
- Presents with constipation and abdominal distension in infancy.

Barium enema and colonoscopy

- Barium enema outlines the lumen and haustral pattern of colon.
- Colonoscopy allows direct visualization, biopsy and removal of polyps.
- Ileocaecal valve and appendicular orifice are important landmarks inside caecum.

Table 10.5 Sites of the carcinoma in colon

Sites of carcinoma	Vessels to be ligated	Segments of colon to be removed
Caecum and ascending colon	Ileocolic, right colic, and right branch of middle colic	Terminal part of the ileum, caecum, ascending colon, hepatic flexure, proximal one-third of transverse colon
Hepatic flexure and right two-third of transverse colon	Ileocolic, right colic, and middle colic	Terminal part of the ileum, caecum, ascending colon, and right two-third of transverse colon
Left one-third of transverse colon and splenic flexure	Middle colic and left colic	From middle of the ascending colon to the sigmoid colon

Table 10.5 - Surgical planning in carcinoma of colon.

The table links the site of carcinoma with vessels to be ligated and the segment of bowel to be removed. In colorectal surgery, the vessel ligated determines the lymphatic territory and therefore the safe margin of bowel resection.

REVISION SHEET

Quick viva answers

Question	Answer
Beginning of large intestine	Caecum at ileocaecal junction
Length of large intestine	About 1.5 m
Three cardinal features	Taeniae coli, appendices epiploicae, haustra
Parts with mesentery	Appendix, transverse colon and sigmoid colon
Commonest position of appendix	Retrocaecal/retrocolic
Artery of appendix	Appendicular artery from ileocolic artery
Surface point for base of appendix	Near McBurney point
Longest part of colon	Transverse colon
Common site of diverticulosis	Sigmoid colon
Critical point of colon	Splenic flexure
Marginal artery of colon	Marginal artery of Drummond
Parasympathetic supply of distal colon	Pelvic splanchnic nerves S2-S4

Last-minute memory hooks

- Taeniae coli converge at the base of the appendix - use them to find the appendix during surgery.
- Appendix pain begins at umbilicus and later shifts to right iliac fossa.
- Proximal two-third of transverse colon is midgut; distal one-third onward is hindgut.
- Splenic flexure is the watershed/critical point between SMA and IMA territories.
- Sigmoid colon is clinically important for diverticulosis and volvulus.

Diagram labelling checklist

- In the large intestine diagram, label caecum, appendix, ascending colon, hepatic flexure, transverse colon, splenic flexure, descending colon, sigmoid colon, rectum and anal canal.
- In the appendix diagram, mark base, body, tip, mesoappendix and appendicular artery.
- In arterial-supply diagrams, show ileocolic, right colic, middle colic, left colic, sigmoid and superior rectal arteries.
- Always add the marginal artery of Drummond when drawing the colonic arterial arcade.

Common exam mistakes

- Do not write that villi are present in the large intestine; they are absent.
- Do not confuse appendices epiploicae with the vermiform appendix.
- Remember that rectum and anal canal do not show the typical cardinal features of colon.
- Do not forget that left colic flexure is higher and more acute than right colic flexure.

How to write a 5-mark answer

- Start with definition and length, then list parts from caecum to anal canal.
- Mention the three cardinal features and explain why they identify colon.
- Add caecum, appendix and colon blood supply with one labelled diagram.
- Finish with two applied points: appendicitis and carcinoma/diverticulosis.

High-yield anatomical logic

- Caecum and appendix belong to midgut, so early appendicular pain is referred to umbilical region.
- Colon above the distal transverse colon follows SMA territory; distal colon follows IMA territory.
- Long mesenteries make transverse and sigmoid colon mobile and clinically prone to volvulus.
- Fixed retroperitoneal ascending and descending colon are commonly mobilized during surgery.

One-paragraph conclusion

The large intestine is best understood as a wider terminal gut segment specialized for water absorption, fecal formation and storage. Its anatomy is identified by taeniae coli, haustra and appendices epiploicae. The caecum and appendix are important because of appendicitis and ileocaecal anatomy, while the colon is clinically important for carcinoma, diverticulosis, volvulus and intestinal obstruction. Arterial supply follows the superior and inferior mesenteric territories, and lymphatic drainage follows the vessels, which is why surgical resection is planned according to vascular and lymphatic fields.

Final revision line

For exam diagrams, remember the sequence: caecum and appendix in the right iliac fossa, ascending colon to hepatic flexure, transverse colon crossing the abdomen, descending colon on the left, sigmoid colon in the pelvis, and rectum/anal canal terminally. Add arterial supply and lymph drainage beside the diagram for a complete answer.